Maximum Mode Timing Diagrams

Q. Discuss the Memory Read and Write timing diagrams for the 8086 microprocessor operating in Maximum Mode. How does it differ from Minimum Mode?

Definition to Remember

In Maxi
`WR'`, t
signals.

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1. The Role of Status Signals

* **Pre-T1 Phase:** Before the T1 clock cycle even begins, the 8086 updates `S2', S1', S0'` to indicate the upcoming cycle (e.g., `101` for Memory Read, `110` for Memory Write).

* The
Read).

2. Memory Read Timing (Maximum Mode)

* **Pre-T1:** S2', S1', S0' = `1, 0, 1` (Memory Read).

* **T1

3. Memory Write Timing (Maximum Mode)

* **Pre-T1:** S2', S1', S0' = `1, 1, 0` (Memory Write).

* **T1

4. Key Differences from Minimum Mode

* **Signal Source:** 8288 generates ALE, DEN, DT/R', not the CPU.

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Must-Write Points (for 10 marks)

1. In Ma

Quick Recall

`Max M
Status g